

Sean P. Florez

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Security Clearance: Top Secret (active); SCI eligible.
Materials R&D | Production Qualification | Technology Transition

Synopsis

Materials science Ph.D. researcher building **manufacturing qualification systems**, **research automation**, and **materials modeling tools**. Experience supporting a new energetic-materials production line, running large computational campaigns, and evaluating technologies for DoD and DOE.

DoD, Defense Industry & National Laboratory Experience

Engineering Contractor, Rocky Mountain Scientific Laboratory

Jun–Aug 2026

Quality system for a client affiliate's new energetic-materials production line

Automation & Robotics embed, Jun–Aug 2026

- Developed lot-release records covering inspections, approval authority, hold criteria, and nonconformance disposition, with item-level traceability to the governing **military specification**.
- Built a controlled, single-source document set with **10 client-format instructions**, **20 test-specific quality-control packets**, manufacturing procedures, and inspections.
- Automated checks for process-route isolation, link validity, terminology, and consistency between released documents and their source.
- Built a program tracker that generates the Kanban board, decision log, requirements traceability matrix, and design-review record.
- Supported quality, packaging, labeling, on-site commissioning, and hazard and failure-mode analysis.

S&T Scouting Intern

The Joint Staff, J7 / Future Technology Office, Aug–Oct 2025

- Produced **eight technical briefs** on propulsion and energy concepts, assessing readiness, risk, and experimentation paths for TRL 3–5 technologies across Service laboratories and DoD R&D organizations.
- Advised on integrating academic research into DoD initiatives, with briefs informing J7 concepts and project proposals.

High-Performance Computing Intern

Air Force Research Laboratory, May–Aug 2025

Materials & Manufacturing Directorate, Wright-Patterson AFB

- Built LAMMPS shear workflows for hydrated Ti_3C_2 MXene bilayers and ran parameter sweeps on **DoD HPC**.
- Quantified hydration-dependent interlayer shear at 103 MPa dry, 8 MPa at quarter-monolayer coverage, and 39 MPa at full coverage.
- Delivered the first MD evidence that hydration tunes MXene interlayer shear strength by an order of magnitude, with stress windows and design guidance for EM-shielding composites.

Technology Commercialization Intern

Idaho National Laboratory (DOE), May–Aug 2024

- Conducted **20 Energy I-Corps interviews**, developed a commercialization plan for a laboratory technology, built AI and natural-language tooling for the technology-transfer team, and supported a DOE proposal awarded **\$2M**.

Materials Modeling Intern

Air Force Research Laboratory, May–Dec 2023

- Automated high-throughput DFT screening of low-work-function emitters for field-emission applications, using Quantum ESPRESSO, ASE, and pymatgen.

Academic Research

Graduate Research Assistant

University of Colorado Boulder, Oct 2024–Present

Heinz Interfaces Laboratory, Prof. Hendrik Heinz

- Ran the production MXene shear campaign of **780 NVT trajectories** on CU's Alpine cluster, zero failures, against a protocol frozen before production and independently audited afterward from raw trajectories. Termination sets interlayer shear resistance across a threefold range ($\tau_{0.5}$ 42–135 MPa).
- Designed and ran the MD campaign behind a lead-author manuscript on N-ethylcarbazole dehydrogenation over Pt nanocrystals for liquid hydrogen carriers; reactive configurations concentrate at low-coordination sites, so local coordination, rather than facet identity, sets where hydrogen release happens.
- Wrote the technical narrative for an ALCF Director's Discretionary award (**~26,150 node-hours** across Polaris, Crux, Sophia, and Graphcore; granted May 2026).
- Parameterized INTERFACE FF models for copper oxides as the first-year deliverable of an **industry-funded force-field program**, calibrated exclusively against experimental data and validated within $\pm 5\%$ of both experimental elastic references.

- Built and open-sourced [OpenSDL](#), an Apache-2.0 framework for computational and autonomous laboratories, built on a durable workflow runtime with policy checks, resource leases, and restart reconciliation over content-addressed artifact storage. ~26k non-test lines of Python, ~600 test functions, CI.

Undergraduate Research Assistant

Hennig Group, University of Florida, Oct 2021–May 2024

- Benchmarked VASPsol against experimental solvation energies; optimized cavity and screening parameters with grid search, Nelder–Mead, and surrogate modeling.

Education

University of Colorado Boulder

Expected May 2028

Ph.D., Materials Science & Engineering. Advisor: Prof. Hendrik Heinz.

University of Florida

May 2024

B.S., Materials Science & Engineering. Minor in Chemistry.

Core Skills

Manufacturing & Quality: Lot-release records; military-specification traceability; requirements traceability matrices; PDR/CDR stage gates; controlled procedures; nonconformance disposition; hazard and failure-mode analysis; bills of materials; vendor qualification.

Research Automation: Campaign orchestration; resource scheduling; adapters for a simulated lab, local compute, and human tasks; run provenance; restart recovery; experiment-selection workflows; policy checks.

Software & HPC: Python; Bash; Linux; Git; Docker; SQLite; pytest; CI; Slurm; PBS Pro; checkpoint/restart; GPU and CPU-MPI workflows.

Materials & Modeling: LAMMPS; NAMD; VASP/VASPsol; Quantum ESPRESSO; ASE; pymatgen; MDAnalysis; force-field parameterization; elastic and surface-energy validation.

ML & Data: scikit-learn; TensorFlow; RDKit; SOAP descriptors; regression and classification; surrogate modeling; uncertainty analysis.

Scale: ~26k node-hours awarded on DOE leadership systems; 780-trajectory audited production campaigns; 10^6 – 10^7 MD timesteps; multi-GB trajectory analysis.

Leadership & Programs

Co-Founder, [LabLink Initiative](#) (nonprofit)

Aug 2024–Present

Partnerships with three Sacramento-area community colleges to move students into DOE Office of Science internships (CCI and SULI), plus a Phi Theta Kappa consortium webinar that drew more than 250 students. Built the matching pipeline and the tooling behind it.

Selected Research Outputs

Co-author: *INTERFACE Force Field for Alumina with Validated Bulk Phases and a pH-Resolved Surface Model Database for Electrolyte and Organic Interfaces*. [arXiv:2601.12570](#), 2026 (preprint).

Lead-author manuscripts in preparation: *Modeling of MXene Shear Properties via Atomistic Molecular Dynamics Simulations*, with AFRL Materials & Manufacturing Directorate co-authors;

Understanding Hydrogen Release Mechanisms of N-Ethylcarbazole on Pt Nano-Crystal Catalysts.